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Predictive Analytics, DDS-855

Assignment 5: Build and Evaluate Classification Models

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Multi-Class Prediction of Obesity Risk: Comparative Evaluation of Regularized Multinomial Logistic Regression, Linear Discriminant Analysis, Naive Bayes, and Support Vector Machine

In this report, I summarize the Kaggle component of Assignment 5 using the attached training and test files, with the goal of predicting the seven-class obesity outcome labeled NObeyesdad. Comparative evaluation across statistical and machine learning classifiers is appropriate in obesity prediction because logistic regression remains a traditional baseline while machine learning methods are increasingly popular, yet recent evidence shows that more complex methods do not automatically outperform logistic regression in this domain (Boakye et al., 2025). That framing is especially relevant in obesity research, where machine learning has already been used to classify and predict obesity-related risks and outcomes from heterogeneous health and behavioral data (DeGregory et al., 2018). Recent obesity-focused work further shows that predictive models can achieve strong discrimination while still supporting interpretation when the analyst evaluates feature importance and model behavior rather than treating the algorithm as a black box (Huang & Huang, 2024).

I analyzed 20,758 labeled observations in the training file and 13,840 unlabeled observations in the test file. The target classes were moderately balanced, ranging from 11.69% for Overweight_Level_I to 19.49% for Obesity_Type_III, which reduced the risk that a single majority class would dominate the evaluation. I detected no missing values in either file, so my preprocessing pipeline emphasized consistency rather than repair. I reserved a stratified 20% holdout sample for internal validation, standardized the numeric predictors, one-hot encoded the categorical predictors, and compared four required models: regularized multinomial logistic regression, linear discriminant analysis, Gaussian naive Bayes, and a polynomial-kernel support vector machine. This design aligns with recent obesity-methods literature that explicitly

compares logistic regression with machine learning approaches and cautions against assuming that a more complex model will necessarily perform better (Boakye et al., 2025). The regularized logistic model also satisfied the assignment requirement for regularized regression.

The holdout results show a clear but not overwhelming leader. The support vector machine achieved the strongest overall performance with an accuracy of .879 and a macro F1 score of .865. Regularized multinomial logistic regression followed closely with an accuracy of .869 and a macro F1 score of .854, while linear discriminant analysis remained respectable at .823 accuracy and .804 macro F1. Naive Bayes declined sharply to .606 accuracy and .544 macro F1. From my perspective, this ranking suggests that the decision boundary in the Kaggle data is partly non-linear and cannot be fully captured by the conditional independence structure assumed by naive Bayes. At the same time, the small gap between the support vector machine and regularized logistic regression indicates that the underlying signal is strong enough that even a more interpretable discriminative model remains highly competitive.

The confusion matrix for the best model helps explain the substantive pattern of error. The support vector machine classified Obesity_Type_II and Obesity_Type_III with very high consistency, whereas the more difficult errors clustered among Normal_Weight, Overweight_Level_I, and Overweight_Level_II. That result is plausible because adjacent weight categories are expected to share overlapping anthropometric and behavioral profiles, while the most extreme obesity states are easier to separate. The regularized multinomial logistic regression output was also informative. Across classes, the largest coefficients were dominated by weight, height, gender, and family-history indicators, which is directionally consistent with the obesity literature reviewed by DeGregory et al. (2018) and with the more recent transparent

obesity-prediction application reported by Huang and Huang (2024). In other words, the Kaggle data behaved in a way that was statistically and substantively believable rather than arbitrary.

My assumption review also aligned with the empirical results. Multinomial logistic regression assumes a stable relationship between predictors and class log-odds, and in this project regularization helped control coefficient instability after one-hot encoding expanded the feature space. Linear discriminant analysis assumes approximate within-class normality and comparable covariance structure across classes, yet my covariance summaries varied meaningfully across obesity levels, which weakens the equal-covariance assumption and likely explains part of LDA's performance loss. Naive Bayes imposes an even stronger conditional independence assumption that is unlikely to hold for obesity-related body measures, habits, and family-history variables, so its weaker performance was expected. The support vector machine required fewer distributional commitments and was therefore better suited to the heterogeneous class boundaries present in this dataset. Overall, I would treat the support vector machine as my strongest submission candidate and the regularized multinomial logistic regression model as my strongest interpretable benchmark. More broadly, that pattern is consistent with recent evidence that machine learning does not automatically outperform logistic regression for obesity risk prediction (Boakye et al., 2025).

Table 1

Holdout Validation Performance by Model

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Support Vector Machine	0.879	0.865	0.866	0.865
Regularized Multinomial Logistic Regression	0.869	0.854	0.855	0.854
Linear Discriminant Analysis	0.823	0.803	0.806	0.804
Naive Bayes	0.606	0.591	0.570	0.544

Note. Scores were computed on the same stratified 20% holdout partition.

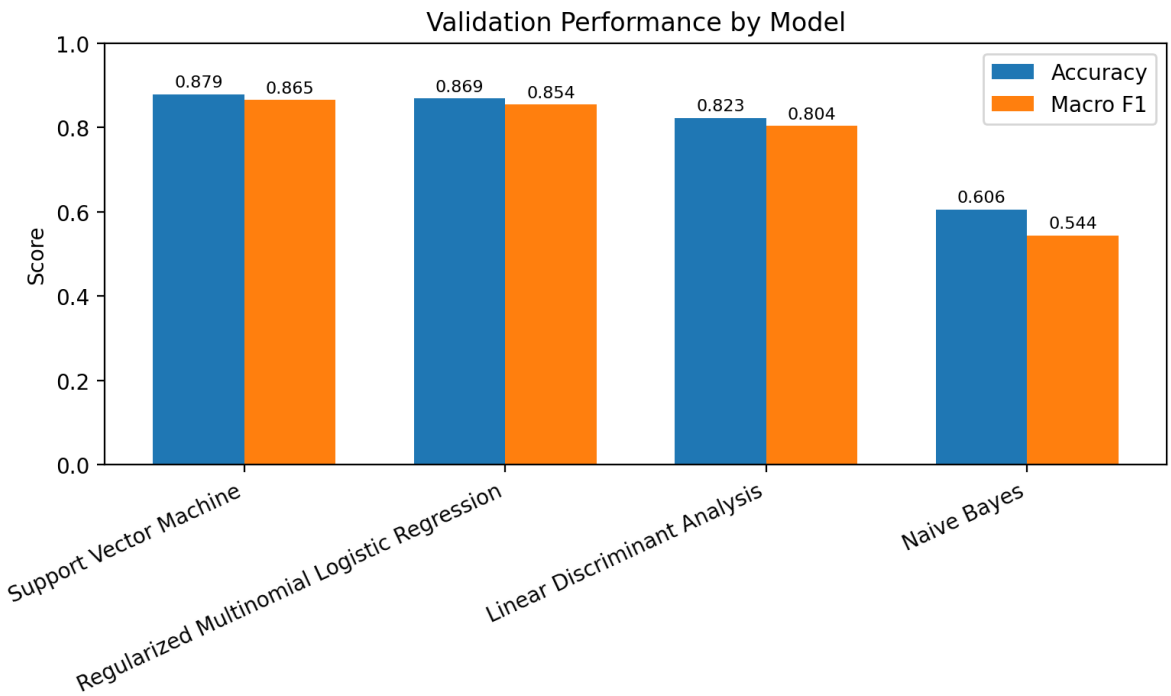
Figure 1

Training Class Distribution

Note. The target classes are reasonably balanced, although Obesity_Type_III is the largest class and Overweight_Level_I is the smallest.

Figure 2

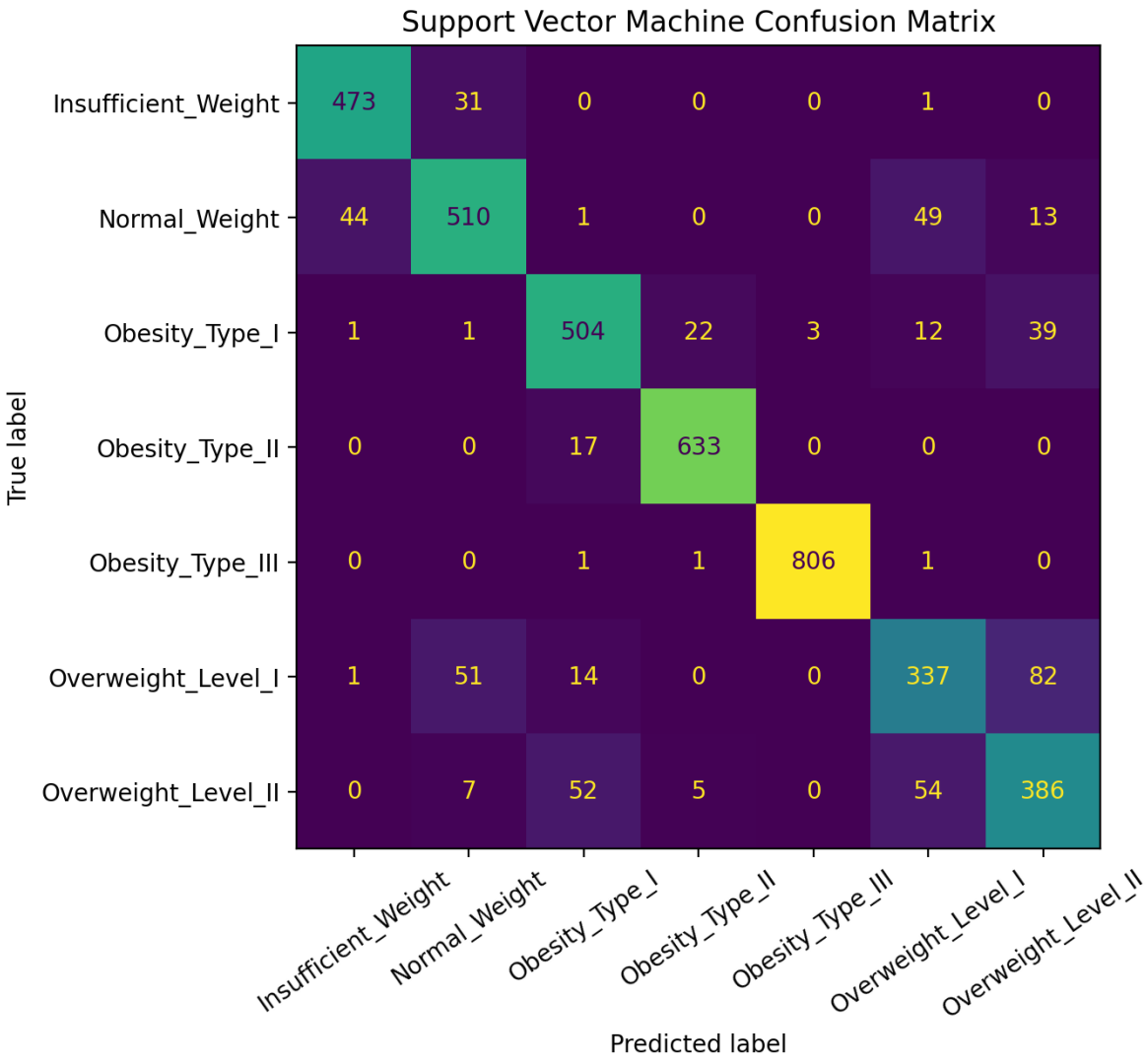
Holdout Validation Performance by Model



Note. The support vector machine achieved the highest accuracy and macro F1, with regularized multinomial logistic regression close behind.

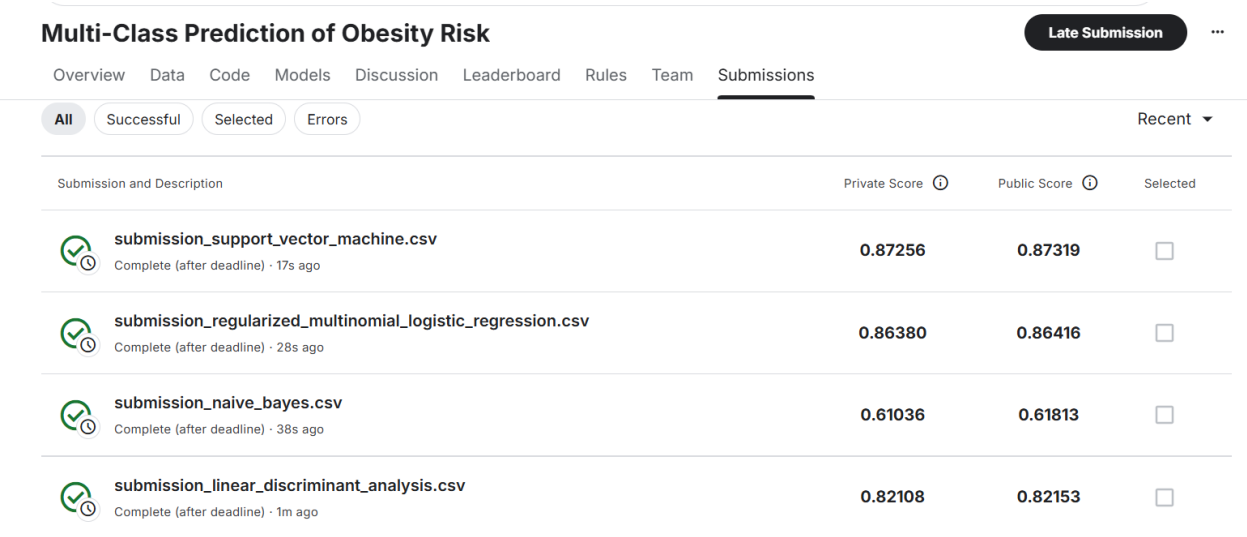
Figure 3

Support Vector Machine Confusion Matrix



Note. Misclassification is concentrated in neighboring middle categories, whereas the more extreme obesity classes are separated more cleanly.

Kaggle Submission Evidence



Multi-Class Prediction of Obesity Risk

Late Submission

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

All Successful Selected Errors

Recent

Submission and Description	Private Score	Public Score	Selected
submission_support_vector_machine.csv Complete (after deadline) · 17s ago	0.87256	0.87319	
submission_regularized_multinomial_logistic_regression.csv Complete (after deadline) · 28s ago	0.86380	0.86416	
submission_naive_bayes.csv Complete (after deadline) · 38s ago	0.61036	0.61813	
submission_linear_discriminant_analysis.csv Complete (after deadline) · 1m ago	0.82108	0.82153	

GitHub Repository Placeholder

https://github.com/National-University-5a26phd1/National_University_PhD_Predictive_Analytics/tree/main/Week_5

References

DeGregory, K. W., Kuiper, P., DeSilvio, T., Pleuss, J. D., Miller, R., Roginski, J. W., Fisher, C. B., Harness, D., Viswanath, S., Heymsfield, S. B., Dungan, I., & Thomas, D. M. (2018). A review of machine learning in obesity. Obesity Reviews, 19(5), 668-685. https://doi.org/10.1111/obr.12667

Huang, A. A., & Huang, S. Y. (2024). Application of a transparent artificial intelligence algorithm for US adults in the obese category of weight. PLOS ONE, 19(5), e0304509. https://doi.org/10.1371/journal.pone.0304509

Boakye, N. F., O'Toole, C. C., Jalali, A., & Hannigan, A. (2025). Comparing logistic regression and machine learning for obesity risk prediction: A systematic review and meta-analysis. International Journal of Medical Informatics, 199, 105887. https://doi.org/10.1016/j.ijmedinf.2025.105887